Multi-Scale Collision Counting for Rényi Entropy Rate Estimation

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***Abstract*—We present a method for estimating the Rényi 2-entropy rate h_2 of stationary ergodic sources from raw byte streams. The method expands bytes to nibbles ($\alpha = 16$), computes debiased collision probabilities $\hat{F}_2(k)$ at k-gram orders $k = 1, \ldots, K$ via the falling-factorial estimator, and extracts the entropy rate from the ordinary least-squares slope of $\log \hat{F}_2(k)$ versus k. On synthetic Markov chains with known ground truth, the estimator achieves median error < 0.003 at $n = 50{,}000$ with $R^2 > 0.999$. On 124 DNS tunnel capture files spanning ten tool families, the Rényi entropy rate h_2 alone (ARI $= 0.720$) outperforms Shannon entropy—both single-scale ($\mathrm{ARI} = 0.246$) and multi-scale slope ($\mathrm{ARI} = 0.204$)—by $3.5\times$ for model-free tool classification. This advantage persists after Miller-Madow debiasing of the Shannon estimator, confirming it is intrinsic to the collision probability functional rather than an artifact of estimation bias. The (h_2, h_3, h_4) multi-order fingerprint provides modest additional gain ($\mathrm{ARI} = 0.747$). We additionally show that R^2 of the linear fit decreases monotonically with Markov order (orders 0–4), serving as a non-parametric memory depth diagnostic. The estimator runs in $O(nK)$ time with $O(\alpha^K)$ space. Code is available at https://github.com/0x1Adi/renyi-entropy-rate-estimation.**

***Index Terms*—Rényi entropy, entropy rate, collision probability, DNS tunnel detection, anomaly classification**

I. Introduction

The collision probability $F_2 = \sum_i p_i^2$, introduced by Friedman [1] as the Index of Coincidence for cryptanalysis of polyalphabetic ciphers, measures the probability that two independently drawn symbols match. It is directly related to Rényi's 2-entropy via $H_2 = -\log_2 F_2$ [2]. Despite its century-old provenance, F_2 remains widely used: in randomness testing [3], privacy accounting [4], and streaming algorithms [5].

For sequential data, the *entropy rate* h captures the per-symbol information content accounting for temporal dependencies. Standard plug-in estimation requires $n \gg \alpha^k$ samples for k-gram frequencies to converge [6]—infeasible at byte resolution ($\alpha = 256$) beyond $k = 1$. We address this by operating at nibble resolution ($\alpha = 16$), enabling $k_{\max} = 3$–5 at typical stream lengths ($n \geq 5{,}000$ bytes).

Our key finding, validated through a controlled 9-method ablation (Table II), is that the Rényi collision functional $F_2 = \sum p_i^2$ captures discriminative structure in sequential data that Shannon entropy $H_1 = -\sum p_i \log p_i$ does not, even when both are computed at identical alphabet size, identical k-gram orders, and with comparable debiasing. This is not obvious *a priori*: both functionals are determined by the same probability vector $\mathbf{p}$.

Contributions. (C1) A debiased multi-scale estimator for h_2 via OLS slope of $\log \hat{F}_2(k)$, empirically validated on Markov chains (Section II). (C2) R^2 of the linear fit as a non-parametric Markov order diagnostic (Section III). (C3) A 9-method ablation isolating the source of improvement for DNS tunnel classification (Section IV).

Related work. Gecchele [7] provides an unbiased single-scale F_2 estimator but does not consider multi-scale slopes or entropy rates. Kontoyiannis et al. [8] give finite-sample bounds for Shannon entropy rate of Markov chains. Yu et al. [9] independently apply multi-order Rényi entropies ($q = 2, 3, 4$) to CAN-bus intrusion detection—the closest methodological antecedent, though in a different domain with different estimation. Skorski [10] provides improved collision estimator analysis. Supervised methods such as GraphTunnel [11] achieve $>98\%$ accuracy on DNS tunnel detection but require labeled training data; our method requires only reference centroids with no model fitting.

II. Method

A. Debiased Collision Probability

Let $\mathbf{c} = (c_1, \ldots, c_d)$ be the vector of k-gram counts from $n_k = n - k + 1$ symbols over alphabet α, where $d = \alpha^k$. The falling-factorial debiased estimator is:

$$\hat{F}_q = \frac{\sum_{i=1}^{d} c_i^{(q)}}{n_k^{(q)}}, \quad x^{(q)} = x(x-1)\cdots(x-q+1) \tag{1}$$

which is unbiased for the multinomial model [7].

Nibble expansion. Each byte is split into two 4-bit nibbles, giving $\alpha = 16$. This yields $k_{\max} = 3$ at $n = 5{,}000$ bytes (10,000 nibbles, since $16^3 = 4{,}096 < 10{,}000/2$) and $k_{\max} = 5$ for $n > 500{,}000$ bytes.

B. Multi-Scale Entropy Rate

For a stationary ergodic source with transition matrix P, the Hadamard square M_2 has entries $(M_2)_{ij} = P_{ij}^2$. For reversible chains:

$$F_2(k) = \boldsymbol{\pi}^\top M_2^k \boldsymbol{\pi} = \sum_j w_j \lambda_j^k \tag{2}$$

TABLE I
R^2 OF $\log \hat{F}_2(k)$ VS. k BY MARKOV ORDER (20 REPLICATES, $\alpha = 4$, $n = 50{,}000$).

Order m	R^2 (mean $\pm$ std)	Monotonic
0	0.9999998 ± 0.0000002	—
1	0.9999995 ± 0.0000013	$\downarrow$
2	0.9999477 ± 0.0000177	$\downarrow$
3	0.9998476 ± 0.0000313	$\downarrow$
4	0.9997505 ± 0.0000356	$\downarrow$
5	0.9997920 ± 0.0000205	$\uparrow$ (uptick)

where $\lambda_1 > |\lambda_2| \geq \cdots$ are eigenvalues of M_2 and $w_j > 0$. For large k, $\log F_2(k) \approx k \log \lambda_1 + \log w_1$, giving:

$$h_2 = -\log_2 \lambda_1(M_2) = -\frac{\text{slope}}{\ln 2} \tag{3}$$

where slope is the OLS coefficient of $\log \hat{F}_2(k)$ versus k.

Adaptive truncation. We set $k_{\max} = \max\{k : \alpha^k/n_k < 0.5\}$.

C. Accuracy and Convergence

We generate sequences from 15 Markov(1) chains at $\alpha \in \{2, 4, 8\}$ with 5 random transition matrices each, at sample sizes $n \in \{5{,}000;\ 20{,}000;\ 50{,}000\}$.

Results. At $\alpha \in \{2, 4\}$: all 30 estimates at $n = 50{,}000$ achieve $|\hat{h}_2 - h_2| \leq 0.025$. At $\alpha = 8$: 4/5 within 0.05; one near-degenerate matrix shows systematic overestimation ($\sim$0.09). Overall median absolute error at $n = 50{,}000$: 0.0024. $R^2 > 0.999$ for all chains.

RMSE $\times \sqrt{n}$ ranges from 0.83 to 1.49 across $n \in [500, 100{,}000]$, with a mild upward trend suggesting a convergence rate slightly slower than $O(1/\sqrt{n})$.

III. R^2 AS MARKOV ORDER DIAGNOSTIC

For an order-m source, $\log F_2(k)$ is exactly linear when dominated by a single eigenvalue, but sub-dominant contributions create deviations at higher orders. We generate 20 replicates of order-m chains ($m = 0, \ldots, 5$) at $\alpha = 4$, $n = 50{,}000$, and compute R^2 of the $\log \hat{F}_2(k)$ linear fit.

R^2 decreases monotonically from order 0 to order 4 (Table I). Order 5 shows a slight uptick (0.99979 vs. 0.99975), which we attribute to $k_{\max}$ truncation limiting visibility of higher-order nonlinearity. Orders 0 and 1 are not separable ($\Delta R^2 < 10^{-6}$). The diagnostic requires high-precision computation to distinguish adjacent orders.

IV. DNS TUNNEL CLASSIFICATION

A. Dataset

We use the GraphTunnel DNS dataset [11]: 124 pcap files spanning 10 families (82 benign, 42 tunnel across 8 tools). DNS query names are extracted via `dpkt` and concatenated into a single byte stream per pcap.

TABLE II
9-METHOD ABLATION ON 124 DNS PCAPS. ALL METHODS USE THE SAME DATA, SAME PCAP PARSING, SAME LOO NEAREST-CENTROID CLASSIFIER. METHODS 3A AND 3B CONFIRM DEBIASING IS NOT THE SOURCE OF RÉNYI'S ADVANTAGE.

#	Method	α	Dim	Type	ARI
1	Shannon H_1 byte	256	1	entropy	0.246
2	Rényi H_2 byte single	256	1	entropy	0.124
3a	Shannon rate (plug-in)	16	1	rate	0.204
3b	Shannon rate (MM debiased)	16	1	rate	0.204
4	**Rényi h_2 slope**	**16**	**1**	**rate**	**0.720**
5	**Rényi (h_2, h_3, h_4) slope**	**16**	**3**	**rate**	**0.747**
6	Shannon 3D	16	3	entropy	0.460
7	Rényi (H_2, H_3, H_4) single	256	3	entropy	0.336

B. Ablation Design

To isolate the source of classification improvement, we compare 9 methods at controlled alphabet, scale, dimensionality, and debiasing conditions (Table II). Classification uses leave-one-out nearest-centroid with Adjusted Rand Index (ARI).

C. Results and Analysis

The ablation reveals three findings:

Rényi beats Shannon at identical conditions. At $\alpha = 16$ with the same multi-scale slope method: Rényi $h_2 = 0.720$ vs. Shannon rate $= 0.204$ (plug-in) and 0.204 (Miller-Madow debiased). The $3.5\times$ advantage is intrinsic to the collision functional, not an estimation artifact.

Multi-scale slope is necessary but not sufficient. The slope method improves Rényi dramatically (byte single $0.124 \rightarrow$ nibble slope 0.720) but hurts Shannon (byte $0.246 \rightarrow$ nibble slope 0.204). The slope captures sequential k-gram structure, but Shannon's logarithmic weighting dilutes the discriminative peaks that $F_2 = \sum p_i^2$ amplifies.

Multi-order adds modest gain. Adding h_3 and h_4 improves ARI from 0.720 to 0.747 ($+3.7\%$). Shannon benefits more from added dimensions ($0.204 \rightarrow 0.460$), suggesting Shannon's weakness is partly compensated by multi-scale diversity.

D. Cross-Domain: EVM Bytecodes

On 161 EVM smart contract bytecodes (118 vulnerable, 43 safe), single-scale Rényi H_2 at $\alpha = 256$ achieves Cohen's $d = 1.32$ and AUC $= 0.818$. The multi-scale approach at $\alpha = 16$ gives weaker results ($d = 0.75$, AUC $= 0.707$) because the discriminative signal for bytecodes is compositional (opcode frequency) rather than sequential.

V. DISCUSSION

Why does F_2 beat Shannon? $F_2 = \sum p_i^2$ weights each probability by itself, amplifying high-probability k-grams. DNS tunnel tools create systematic encoding patterns (base32, base64, hex) that produce a few dominant nibble k-grams. Shannon's $-p \log p$ weighting distributes emphasis more uniformly, diluting these peaks. The multi-scale slope then converts the amplified F_2 signal into an entropy rate that captures transition-level encoding structure.

Compositional vs. sequential signals. The choice of method depends on the data domain. For DNS (short strings, encoding structure): $\alpha = 16$ slope captures sequential nibble transitions. For EVM bytecodes (long streams, opcode distribution): $\alpha = 256$ single-scale captures compositional frequency. This suggests practitioners should select α and the slope/single-scale mode based on whether the discriminative signal is sequential or compositional.

Limitations. (1) The stationarity assumption is violated by real DNS streams with query bursts; the estimator averages over non-stationary segments without detecting regime changes. (2) R^2 cannot separate Markov order 0 from 1, and the uptick at order 5 limits the diagnostic to orders 0–4. (3) The method requires reference centroids from labeled data for classification; it is model-free but not label-free. (4) We validate on two real-world domains (DNS tunnels and EVM bytecodes). System binaries ($n = 93$), text files ($n = 2$), and crypto streams ($n = 8$ synthetic) are illustrative only and do not constitute validated applications.

VI. Conclusion

The OLS slope of debiased $\log \hat{F}_q(k)$ across k-gram orders provides a practical $O(nK)$-time estimator for Rényi entropy rates, validated on synthetic Markov chains (median error <0.003). A controlled 9-method ablation confirms that the Rényi collision functional captures discriminative sequential structure that Shannon entropy does not, achieving $3.5\times$ the classification performance at identical alphabet, scale, and debiasing conditions on DNS tunnel data. Code is available at https://github.com/0x1Adi/renyi-entropy-rate-estimation.

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